a

The analysis unit is a subtype-defined developmental neighborhood

CM identity

Ankrd1; working; Slit2; Ddc; cycling

Maturation within stage

Continuous21-gene score and separate functional-RNA components

Non-CM subtypes

Postn / Gfpt2 / Col8a1 FBs; capillary / endocardial / arterial ECs

Broad lineage context is background; it is not a limit on subtype-defined analyses

b

Component-specific candidates from the observed data

P7 · cycling CM + Postn FB

Lower maturation/metabolic-regulatory RNA; annotation and radius sensitivity remain

P14 · Ankrd1-CM + endocardial EC

Higher sarcomere RNA in local models; not a uniform increase in all maturation components

c

Choose subtype-matched functional and lineage experiments

Fibroblast comparison

Postn versus Gfpt2 context around early cycling CMs; assess metabolism, matrix and completed division

EC × CM identity

Endocardial-like versus capillary ECs around Ankrd1 versus working CMs; sarcomeres, calcium and force

State continuity

Trace validated early FB labels into later matrix-specialized states; do not treat OT as ancestry